WORKSHEET · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 4: Swapping and the Bubble Pass

Module 2 | 45 minutes | Swapping and comparisons | Grades 6–8 · ages 12–14

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

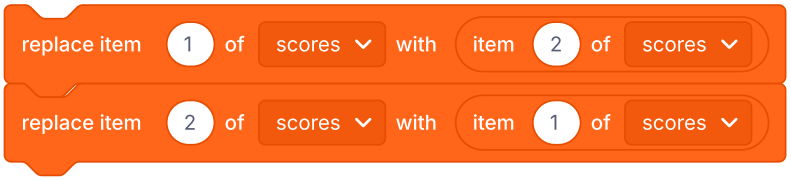
a. swap ___ A. Next to each other.
b. temporary variable ___ B. One trip through the list comparing neighbours.
c. pass ___ C. A variable that holds a value briefly.
d. adjacent ___ D. Exchange two values.

2. Plan your code

Before you build it: Safe swap: scores holds 5, 3, 8, 1. Swap item 1 and item 2 using a temp variable, so the list becomes 3, 5, 8, 1. Write or draw the blocks you will use.

3. Debug the swap

This swap gives 3, 3, 8, 1 instead of 3, 5, 8, 1. Why? Circle one.



a. A. Item 1 was overwritten before it was copied — it needs a temp variable
b. B. The items should be swapped in reverse order
c. C. replace item doesn't work with numbers

4. Plan your code

Before you build it: One bubble pass: swap (i) is defined for you. Set j to 1 and repeat (length of scores − 1): if item j > item j + 1, call swap (j); change j by 1. The list 5, 3, 8, 1 should become 3, 5, 1, 8. Write or draw the blocks you will use.

5. Explain your thinking

Explain why a swap needs a temporary variable. Use the juice-cup idea or your own example.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

- ☐ I can swap two items with a temp variable.
- ☐ I can write one bubble pass.
- ☐ I can explain what a pass achieves.